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Effect of harsh environmental conditions on the tensile properties of GFRP bars

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ABSTRACT

This paper presents the test results of an experimental study carried out to investigate the residual tensile properties of new generation of GFRP bars after being subjected to several harsh environmental conditions for different periods. The bars were exposed to ten different environmental conditions for 6, 12, and 18 months. In addition to a control laboratory condition, the environments included exposure to ordinary tap water and sea water at two temperatures (room and 50 °C), sea water dry/wet, alkaline solution, and hot-dry condition at 50 °C. Furthermore, the environments included two typical harsh field conditions of the Kingdom of Saudi Arabia. The performance of the GFRP bars was evaluated by conducting tensile tests on the bars after different exposure periods. Scanning electron microscope was used to investigate the degradation mechanism of the bars. After 18 months of exposure, the test results showed that at 50 °C the tap water and alkaline solution had the maximum harmful effect on the tensile strength of the tested GFRP bars. The two field conditions considered in the present study did not show any significant effect on the tensile properties of the bars. The test results also showed that the GFRP bars tested in this study had better resistance to alkalinity and water compared to most of the GFRP bars in the literature.

1. Introduction

For the last two decades, fiber reinforced polymer (FRP) bars have been used in several concrete structures throughout the world due to their several advantages such at corrosion resistance, light-weight, and high strength. Among different types of FRPs, glass FRP (GFRP) bars have drawn more attention in civil engineering applications due to their low cost compared to other types of FRPs such as Carbon or Aramid [1–5]. However, some doubt about their durability in concrete environment is still an obstacle against their wide and broad use. Although GFRP reinforcing bars do not exhibit corrosion like steel bars, many studies have reported deteriorations that result in a significant reduction in the tensile strength of GFRP bars when exposed to various solution types such as water and alkaline solutions [1,6–12].

Contrary to the corrosion of steel reinforcement which gives visible warning signs such as cracking or spalling of the concrete, the deterioration of glass fibers in GFRP bars does not result in a product of increased volume and does not lead to cracking and/or spalling of the concrete cover. Although this looks beneficial; no warning sign is a disadvantage. If the tensile properties of the GFRP bars are decreasing as a function of time without visible warning signs, the GFRP-reinforced concrete structure could be considered as having sufficient capacity when in fact it does not [13]. Thus, it is imperative that the reduction in the capacity of GFRP bars under different environmental conditions be properly characterized and accounted for.

Several research studies have been conducted to investigate the durability of GFRP bars under different environmental conditions. Table 1 summarizes the test results of some of these studies. Porter and Barnes [6] conducted accelerated tests on three types of GFRP bare bars in an alkaline solution at a high temperature (60 °C) for periods of 60–90 days. Their tensile tests resulted in residual strengths of 34%, 52%, and 71% compared to the original tensile strength. Alsayed and Alhozaimy [14] and Alsayed et al. [1] investigated the residual tensile strength of GFRP bare bars after being subjected to different environmental conditions for 6 months. The environments included immersion in tap water, sea water and alkaline solution at different temperatures (23, 35, and 50 °C). The reduction in the tensile strength after 6 months in the alkaline solution at 50 °C was about 13.7%. Gaona [15] also conducted a durability study on GFRP bars and measured losses up to 24% for bars conditioned in an alkaline solution at a temperature of 35 °C for 50 weeks. The elastic modulus of the bars, however, increased by 9% for the 50-week conditioning. Chu et al. [7] investigated the effects of moisture and alkalinity on the tensile properties of E-glass/vinylester composite strips. Specimens were placed into deionized water or alkaline solution at different temperatures (23, 40, 60, and 80 °C) for 18 months. The results showed that the tensile strength losses ranged between 35% and 72% of the initial strength. They concluded that there were only slight

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Table 1
Tensile strength of GFRP bare bars exposed to different solutions in various studies.

Reference	Matrix glass material[a]	Bar diameter (mm)	Conditioning solution	Temperature (°C)	Exposure period (days)	Tensile strength loss (%)
Porter and Barnes [6]	E-glass/IP, BV	9.4	Alkaline solution	60	19–81	29–66
Gaona [15]	E-glass/vinylester	16	Alkaline solution (pH = 12)	35	350	23
Chu et al. [7]	E-glass/vinylester	Strip 0.063 × 15.24 mm	Deionized water	23, 40, 60, 80	525	35, 49, 63, 72
			Alkaline solution			42, 47, 61, 62
Al-Zahrani [8]	E-glass/modified vinylester	12	Alkaline solution (pH = 13.5)	60	360	77
			Alkaline + sea water	60		58
			Alkaline + sabkha	60		49
			Thermal variations (22 °C and 60 °C)	22 to 60		5
			Out-door	Varied		6
	E-glass/vinylester	12	Alkaline solution (pH = 13.5)	60	360	71
			Alkaline + sea water			77
			Alkaline + sabkha			66
	E-glass/polyurethane (Thermoplastic)	12	Alkaline solution (pH = 13.5)	60	360	21
			Alkaline + sea water	60		19
			Alkaline + sabkha	60		27
			Thermal variations (22 °C and 60 °C)	22–60		15
			Out-door	Varied		21
Alsayed et al. [1]	E-glass/urethane-modified vinylester	9.5	Water	50	180	2.7
			Alkali solution			13.7
Chen et al. [9]	E-glass/vinylester	9.5	Water	20, 40, 60	120 for 20 °C and 70 for 40 and 60 °C	5, 3, 29
			Alkaline (pH = 13.6)			14, 11, 36
			Alkaline (pH = 12.7)			8, 8, 27
			Sea water			3, 2, 26
	E2-glass/vinylester	9.5	Alkaline (pH = 13.6)	40, 60	60	31, 48
				20	120	45
Kim et al. [11]	E-glass/vinylester	12.7	Water	25, 40, 80	132	11, 16, 22
			Sea water	25, 40, 80	132	14, 13, 19
			Alkaline solution (pH = 13)	25, 40, 80	60	32, 30, 40
	E-glass/modified vinylester	12.7	Water	25, 40, 80	132	12, 19, 56
			Sea water	25, 40, 80	132	19, 18, 57
			Alkaline solution (pH = 13)	25, 40, 80	60	17, 18, 50

[a] IP = isophthalic polyester; BV = bisphenol vinylester.

differences in strength between the deionized water and alkaline environments after the eighteen months of exposure.

Wang [16] conducted several accelerated aging tests on three different diameters of E-glass/vinylester GFRP bars. The bars were conditioned in simulated pore water solution (pH 12.6–12.8) and distilled water at 23, 40 and 60 °C for 150 and 300 days. Losses in the tensile strength of 51, 25 and 32% after 300 days aging in alkaline solution were recorded at 60, 40 and 23 °C, respectively. Al-Zahrani [8] investigated the residual tensile strength of three types of GFRP bars in four aggressive solutions (alkaline, alkaline + seawater, alkaline + sabkha, and acidic) at three different temperatures (60 °C, 24 h thermal variations of 22 °C and 60 °C, and out-door) for 3, 6, 9, and 12 month periods. The tensile test results showed major reduction in the tensile strength of the three types of GFRP bars due to immersion in alkaline environment and sabkha at 60 °C with a maximum reduction in the tensile strength of 27–77%. For thermal variation and out-door, the reduction ranged between 5% and 21%. Won et al. [10] examined the accelerated degradation of E-glass fiber/vinylester resin composite rebar in alkaline solution at 12.6 pH and water. The tensile strength of the GFRP rebar exposed to alkaline environment was found to have markedly reduced with the increase in the exposure duration. The mechanical performance of rebars was also reported to have decreased with the increase in exposure time and temperature because of the increase in the pore distribution.

Another durability study on GFRP bars was carried out by Chen et al. [9]. The bars were exposed to five different solutions: water, two types of simulated alkaline pore solutions of normal and high performance concrete, saline solution, and combined alkaline solution with chloride ions. The aging was accelerated by using elevated temperatures. Wetting and drying and freezing and thawing cycles were combined with some solutions to simulate the coupling effects as expected in field conditions. The results showed that significant strength loss resulted from the accelerated exposure especially for solutions at 60 °C. Continuous immersion resulted in greater degradation than exposure to wetting and drying cycling. In contrast, freezing and thawing cycling combined with solutions had little degradation effects on the GFRP bars. Kim et al. [11] conducted a short-term durability test on two types of commercially available GFRP bars (E-glass/vinylester) under four different environmental conditions (moisture, chloride, alkali, and freeze–thaw cycling) for up to 132 days. In addition to the room temperature, elevated temperatures of 40 and 80 °C were used to accelerate the degradation of the GFRP bars. The results indicated that the strength was reduced in all cases as the immersion time and temperature increased. They also concluded that alkaline environmental condition had more influence on the degradation of GFRP bars than the other influencing factors.

Thus, based on the above review, it can be concluded that there are wide and significant variations in the strength reduction of GFRP bars due to environmental exposure. In addition, most of these studies have been conducted on old generations of GFRP bars. The manufactures are now claiming that their newly developed GFRP bars are much better and possess improved resistance to different environmental conditions especially water and alkaline solutions. Therefore, this study aims at investigating the durability of newly developed GFRP bars under several laboratory and field environmental conditions. The laboratory environments include exposure to ordinary tap water and sea water at two different temperatures. They also include exposure to sea water dry-wet, alkaline solution, and hot-dry condition. The field conditions include two harsh field conditions of the Middle East, the Arabian Gulf area in particular (hot-dry and hot-humid field conditions). The residual tensile strengths of the newly developed GFRP bars considered in this study were also compared with those of several GFRP bars in the literature. In addition, a comparison between the effects of accelerated laboratory and field conditions is also presented. Scanning electron microscope (SEM) was also used to investigate the degradation mechanism of the bars.

2. Experimental program

2.1. GFRP bars

The GFRP bars used in this investigation were chosen based on screening tensile tests carried out on four different types of GFRP bars (E-glass/vinylester) available in the market from different manufacturers. The type with the best stable mechanical properties was taken up for the present investigation. The GFRP bars chosen were 12 mm diameter bars with a special surface profile (ribs) to enhance the bond between bars and concrete. The bars were made of continuous longitudinal glass fibers impregnated in a thermosetting vinylester resin with a fiber content of 83%. Table 2 shows the tensile test results of the GFRP bars used in this study where high tensile strength and modulus were recorded. The tensile properties were also consistent with low standard deviations and coefficients of variation. Fig. 1 shows a picture of these bars.

Table 2
Mechanical properties of the GFRP bars used.

FRP chosen bars		Ultimate strength (MPa)	Modulus of elasticity (GPa)	Fracture strain (%)
ϕ; 12 mm (Area = 113 mm^2)	Average	1478	60.4	2.45
	Standard deviation	29.9	1.69	0.05
	Coefficient of variation (%)	2.03	2.79	2.15

Fig. 1. GFRP bars used in this study.

Table 3
Test matrix.

Environment	Temperature (°C)	Name	No. of specimens
Lab environment			
Unconditioned control in the laboratory	Room	LE	5
Lab conditioned specimens			
Tap water (immersed)	Room	TWR	5
	50	TW50	5
Sea water (immersed)	Room	SWR	5
	50	SW50	5
Sea water (wet/dry)	50	SW50DW	5
Alkaline	50	ALK	5
(hot/dry)	50	HD50	5
Field specimens			
Riyadh area	Field	RF	5
Gulf area (Jubail)	Field	JF	5
Subtotal-1			50
No. of exposure periods			3
Total no. of GFRP bar specimens			150

2.2. Test specimens

A total of 150 GFRP bars were subjected to ten environmental conditions for 6, 12, and 18 months. The environments included exposure to ordinary tap water and sea water at two temperatures (room and 50 °C); sea water dry-wet cycles; alkaline solution; and hot-dry condition at 50 °C. The sea water was brought from the Arabian Gulf-Eastern Province of Saudi Arabia. Alkaline solution (with a pH value of 12.5–13) was prepared using Calcium Hydroxide, Potassium Hydroxide and Sodium Hydroxide. The environments also included two typical field conditions of the Kingdom of Saudi Arabia (Riyadh area and Gulf area). Table 3 summarizes the test matrix and the number of tested GFRP bars. The environmental conditions were divided into three main categories as follows:

2.2.1. Unconditioned specimens

Unconditioned laboratory specimens, LE set, were exposed to a controlled laboratory environment (temperature = 23 ± 2 °C). After 6, 12 and 18 months of exposure, bars were tested in tension according to the ACI 440 Guide [17].

2.2.2. Conditioned specimens in laboratory

In this category, the GFRP bars were exposed to different types of accelerated environmental conditions for 6, 12, and 18 months. After each exposure period, the specimens were taken out from the environments and kept in the lab environment for about a week before testing. The effect of different environments on the tensile properties of the bars was determined by comparing the test results with control. The accelerated environmental conditions considered in this study were:

- Immersion in tap water at ambient and 50 °C (TWR and TW50 specimens).
- Immersion in sea water at ambient and 50 °C (SWR and SW50 specimens).
- Wet-dry cycles in sea water at 50 °C (SW50DW specimens).
- Immersion in alkaline solution (pH 12.5–13) at 50 °C (ALK50 specimens).
- Hot-dry laboratory condition at 50 °C (HD50 specimens).

2.2.3. Field conditions

Two harsh field conditions were considered in this investigation. A set of specimens was exposed to Riyadh hot-dry field conditions, RF set, (representing hot-dry arid land of the Middle East). A second set was exposed to the Eastern coast of the Kingdom at the Gulf area (Jubail city), JF set, which represents the hot-humid environment of the Middle East. It should be mentioned that the monthly average temperature of the two field conditions is almost similar and ranges between 9 and 45 °C with annual average high and low temperatures of 33 and 19 °C, respectively. The annual average relative humidity is about 26% and 52% in Riyadh and Jubail areas, respectively [18].

2.3. Microstructural analysis

Scanning electron microscope (SEM) was used to investigate the phenomena of degradation occurring during aging. The cross sections of the GFRP bars in ALK50 and TW50 environments after 18 months of exposure were examined using the SEM technique and compared to those of the control specimens. The outer parts (close to surface) of the bar samples were selected for examination since they are the portion most subjected and affected by the aggressive exposures.

3. Test results

3.1. Tensile strength, modulus and strain at failure

Table 4 summarizes the average measured tensile strength, modulus and strain at failure of all tested bars. The specimens in the controlled lab environment (LE) almost did not show any degradation in the tensile strength, modulus, or fracture strain. The test results of each set of bars were consistent. The maximum computed coefficients of variation of tensile strength, modulus of elasticity, and strain at failure were 5.6%, 4.3%, and 5.5%, respectively.

Fig. 2. A close view of the typical mode of failure of the tested bars.

3.2. Stress–strain curves and mode of failure

The stress–strain relationships of the tested GFRP bars were almost linear up to failure for all tested bars regardless of the environment or period of exposure. All tested bars had a similar mode of failure which was a brittle fracture with delamination and/or shearing of the matrix. Fig. 2 shows a picture of this typical mode of failure of all tested bars.

4. Analysis and discussion of test results

4.1. Tensile strength

Figs. 3 and 4 show the average tensile strength retention and losses, respectively, of the specimens as a function of exposure

Table 4
Test results.[a]

Exposure period	Environments	Tensile strength		Tensile modulus		Rupture strain	
		Average (GPa)	CV (%)	Average (MPa)	CV (%)	Average (%)	CV (%)
6 months	LE	1482	0.7	60.9	3.5	2.43	2.8
	TWR	1442	1.8	58.9	1.9	2.45	3.4
	TW50	1225	3.0	57.9	0.8	2.12	3.0
	SWR	1447	3.1	60.9	2.8	2.38	5.2
	SW50	1320	1.3	58.6	1.9	2.25	3.2
	SW50DW	1338	3.0	61.2	0.9	2.19	3.8
	ALK50	1262	2.5	58.2	3.3	2.17	5.3
	HD50	1478	0.6	59.9	2.6	2.47	3.2
	RF	1477	2.6	59.1	2.5	2.50	3.0
	JF	1479	2.1	60.2	3.1	2.46	1.7
12 months	LE	1469	2.9	59.9	2.7	2.46	5.5
	TWR	1443	2.8	59.6	4.3	2.42	1.5
	TW50	1148	1.9	60.2	0.1	1.92	2.2
	SWR	1338	1.3	57.6	3.4	2.33	2.7
	SW50	1352	2.4	58.5	3.8	2.36	5.3
	SW50DW	1348	2.4	59.1	2.7	2.28	1.0
	ALK50	1180	5.6	58.7	1.4	1.99	5.3
	HD50	1480	0.8	57.8	4.3	2.57	4.2
	RF	1463	2.1	59.7	2.2	2.45	0.1
	JF	1461	1.3	58.6	1.0	2.49	2.2
18 months	LE	1485	0.6	60.1	1.4	2.47	1.1
	TWR	1406	3.8	58.0	3.7	2.43	2.8
	TW50	1122	3.0	58.6	2.5	1.92	4.3
	SWR	1326	2.8	58.7	2.3	2.24	0.1
	SW50	1295	4.3	58.6	3.1	2.25	3.2
	SW50DW	1340	0.5	58.9	1.8	2.30	0.8
	ALK50	1128	2.9	59.6	0.6	1.89	3.7
	HD50	1441	3.8	60.3	0.6	2.39	4.1
	RF	1477	0.6	59.9	2.9	2.47	3.1
	JF	1477	1.2	59.5	2.0	2.48	2.0

[a] CV: Coefficient of variation (%) = (standard deviation/average) × 100.

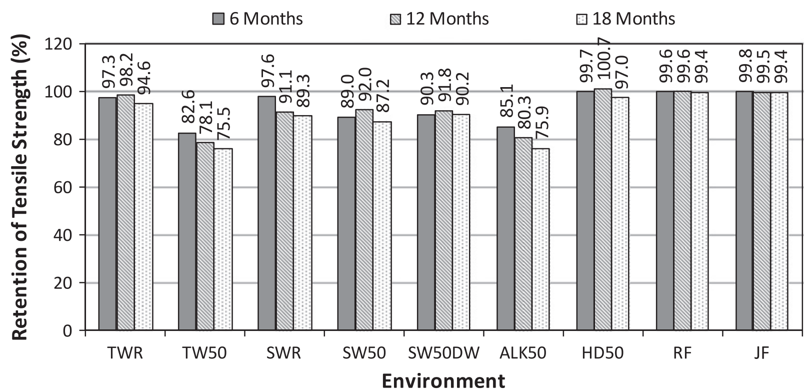


Fig. 3. Retention of tensile strength after 6, 12, and 18 months in different environments.

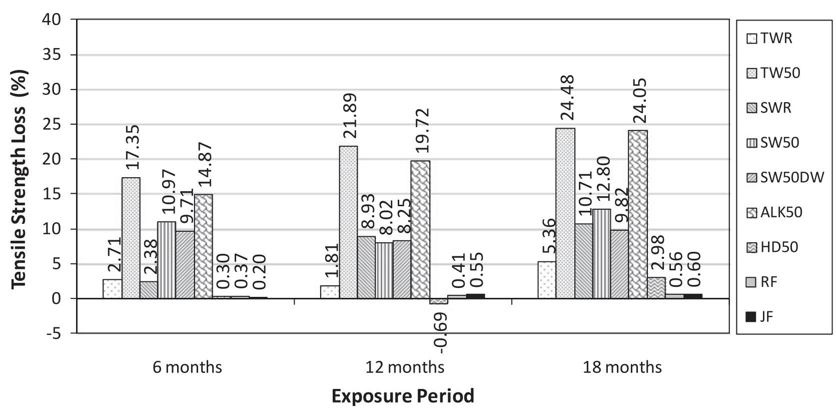


Fig. 4. Tensile strength loss of GFRP bars after 6, 12, and 18 months in different environments.

period and environmental condition. In Fig. 3, the average tensile strength of the conditioned specimen is divided by that of the unconditioned specimen in the LE environment and the corresponding value is denoted as the retention ratio in percent.

For the test specimens in tap water at room temperature, TWR, a slight reduction in the tensile strength was recorded after 6, 12, and 18 months of exposure. The residual strengths were about 97.3%, 98.2%, and 94.6% (losses of 2.7%, 1.8%, and 5.4%), respectively. Fig. 5 shows a comparison between the tensile strength retention of the TWR specimens and other GFRP bars in the literature exposed to a similar condition. It can be noticed that the GFRP bars tested in this study show higher residual strengths compared to most of the GFRP bars in the literature.

For the TW50 specimens, the strength loss was 17.3% after 6 months of exposure. After 12 and 18 months of exposure, this strength loss slightly increased to 21.9% and 24.5%, respectively, which indicates that most of the losses (about 70% of the total loss) occurred in the first 6 months of exposure. Compared to the TWR specimens, it can be noticed that increasing the temperature to 50 °C resulted in a faster degradation in the bars leading to that decrease in the tensile strength. This observation is consistent with studies in the literature which concluded that moisture and temperature are of the main parameters that affect the durability of composite materials. It is well known that moisture absorbed by the composites combined with the temperature of exposure induces stresses in the material which consequently damage fibers,

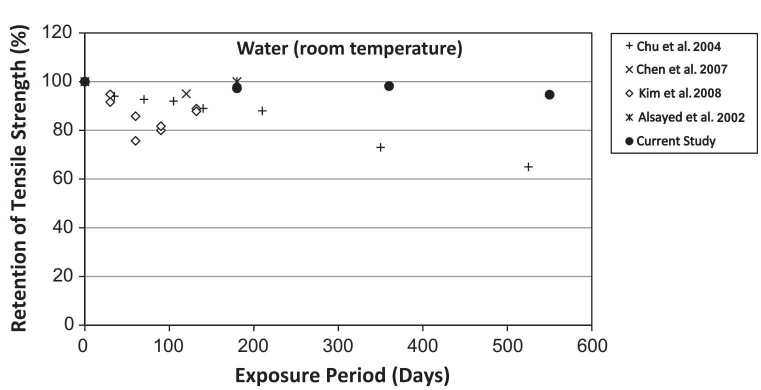


Fig. 5. Retention of tensile strength of GFRP bars exposed to water at room temperature.

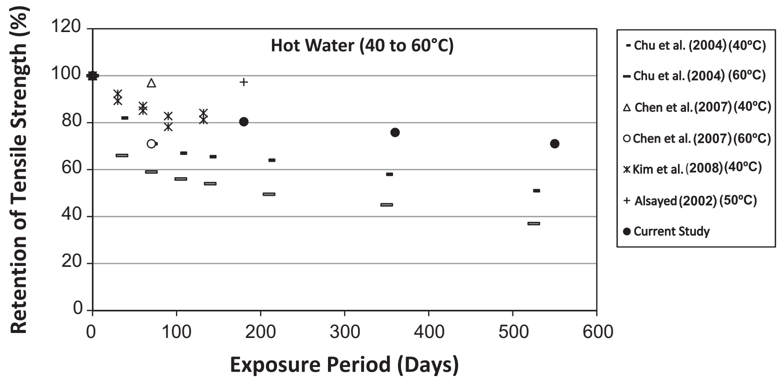


Fig. 6. Retention of tensile strength of GFRP bars exposed to hot water.

matrix, and their interface and decreases the strength of FRP material with time [19–21]. Moisture can also act as a plasticizer disrupting Van-der-Waals bonds in polymer chains [22] and producing fiber–matrix debonding [23]. This phenomenon was observed and explained in section 4.4 (microstructural analysis results) of this paper.

A comparison between the TW50 specimens and other GFRP bars exposed to hot water (40–60 °C) in the literature is presented in Fig. 6. It can be noticed that, except for the two results reported by Alsayed et al. [1] and Chen et al. [9], the GFRP bars investigated in this study had better residual tensile strength compared to all other GFRP bars reported in the literature.

For the specimens in seawater at room temperature, SWR, Fig. 3 shows that the residual tensile strength decreased with time. It was 97.6%, 91.1%, and 89.3%, after 6, 12, and 18 months of exposure respectively which is comparable but slightly less than those recorded in the TWR environment. Increasing the temperature to 50 °C, (SW50 specimens), resulted in additional reduction in the tensile strength after 6 months of exposure. This reduction, however, did not increase with time. After 6, 12, and 18 months, the strength loss was about 11.0%, 8.0%, and 12.8%, respectively. Fig. 7 shows a comparison between the tensile strength retention of the SWR and SW50 specimens and other GFRP bars in the literature exposed to sea water. Again, it can be seen that the GFRP bars considered in this study have better residual tensile strengths compared to their counterpart GFRP bars reported in the literature.

For the specimens in dry-wet seawater at 50 °C, SW50DW, the recorded strengths losses were almost similar to those obtained in the SW50 specimens. After 6, 12, and 18 months of exposure, the average strength loss was about 9.7%, 8.2%, and 9.8%, respectively.

For both SW50 and SW50DW conditions, it can be noticed that most of the tensile strength losses were recorded after the first 6 months of exposure and almost no additional losses were recorded after 12 and 18 months. This could be attributed to the formation of a thin layer of salt on the bars, especially at higher temperature, which decreased the diffusion rate of the solution into the bars.

For the specimens in alkaline solution at 50 °C, ALK50, the performance was almost similar to that of the TW50 specimens. A significant reduction in the tensile strength (about 14.87%) was recorded after the first six months of exposure. After 12 and 18 months of exposure, the strength loss increased to 19.72%, and 24.04%, respectively. This strength loss with time in alkaline exposure was expected. Because of the chemical attack on the glass fibers and the concentration and growth of hydration products between individual filaments, the strength and stiffness of FRP materials may reduce significantly in alkaline environments. A comparison between the ALK50 specimens and other GFRP bars in the literature exposed to alkaline solutions with respect to residual tensile strength is given in Fig. 8. It can be again noticed that the GFRP bars investigated in this study had greater residual tensile strength compared to the GFRP bars in the literature.

For the specimens in the hot-dry laboratory environment, HD50, and the two field conditions; Riyadh field condition, RF, and Gulf field condition, JF, almost no reduction in the tensile strength was recorded after 18 months of exposure. This indicates that the accelerated laboratory environments were too harsh compared to the field conditions considered in the present study.

As a general observation, it can be noticed that the maximum reductions in the tensile strength were recorded in the TW50 and ALK50 environments which is in good agreement with several

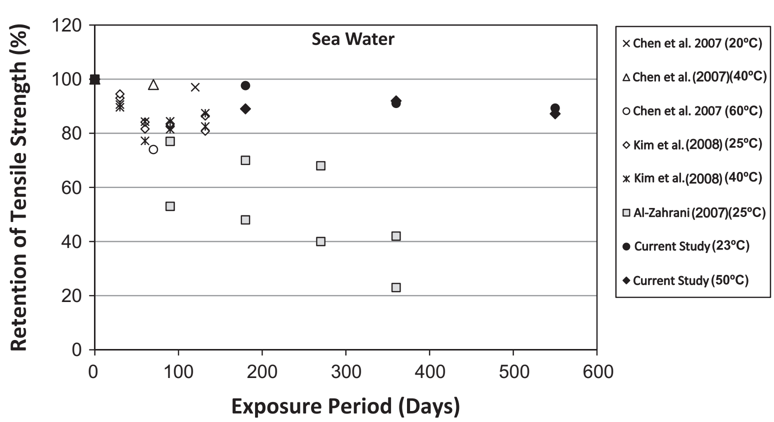


Fig. 7. Retention of tensile strength of GFRP bars exposed to sea water.

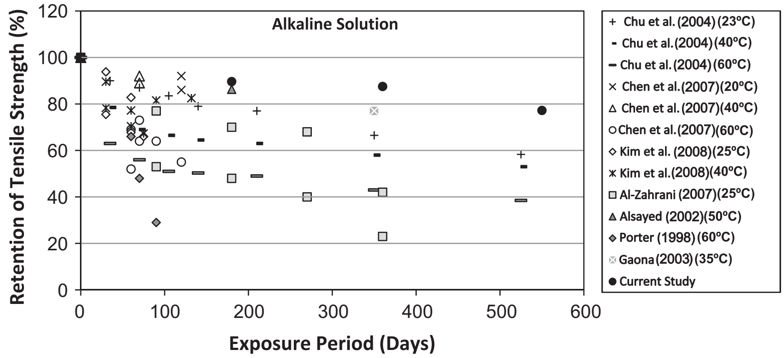


Fig. 8. Retention of tensile strength of GFRP bars exposed to alkaline solution.

researches. Most of the studies in the literature [1,6–9,11,16,17,19,24,25] concluded that moisture and alkalinity are the two main environments that affect the durability of composite materials. However, the current test results showed better performance of the newly developed GFRP bars compared to the earlier bars tested in various studies reported in the literature. This shows the improvement in the manufacturing process of GFRP bars and the promising future of the new generations of GFRP bars in concrete structures.

4.2. Tensile modulus of elasticity

Fig. 9 shows the average residual tensile modulus of the tested GFRP bars after 6, 12, and 18 months in different exposures. It can be noticed that the environmental conditions did not have significant influence on the tensile modulus of GFRP bars. For all environments, a slight decrease ranging between 0% and 5% was observed. These results are in good agreement with the experiments performed by different researchers [9,11,19,24–27].

4.3. Strain at failure

Fig. 10 shows the retention of rupture strain of the tested GFRP bars after 6, 12, and 18 months in different exposures. The rupture tensile strains showed similar pattern to those of the tensile strengths. For the specimens exposed to TWR, HD50, RF, and JF environmental conditions, almost no decrease in the strain at failure was recorded after 18 months of exposure. After 18 months of exposure, the specimens in the SWR, SW50 and SW50DW environments showed a decrease in the strain at failure of about 9.3%, 8.9%,

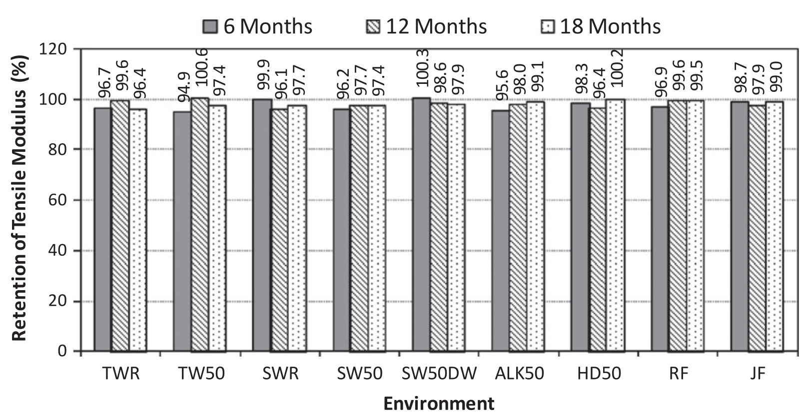


Fig. 9. Retention of tensile modulus of GFRP bars after 6, 12, and 18 months in different environments.

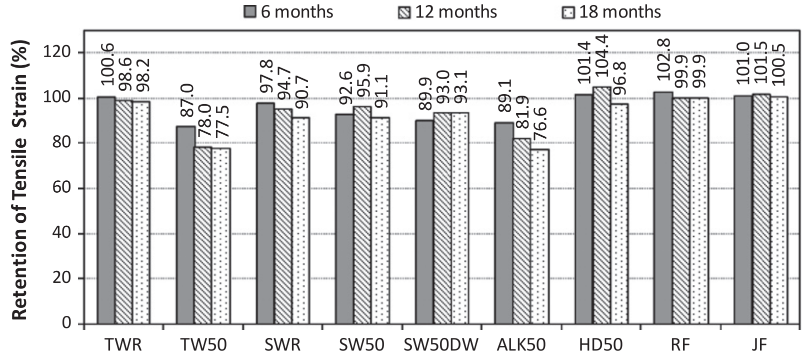
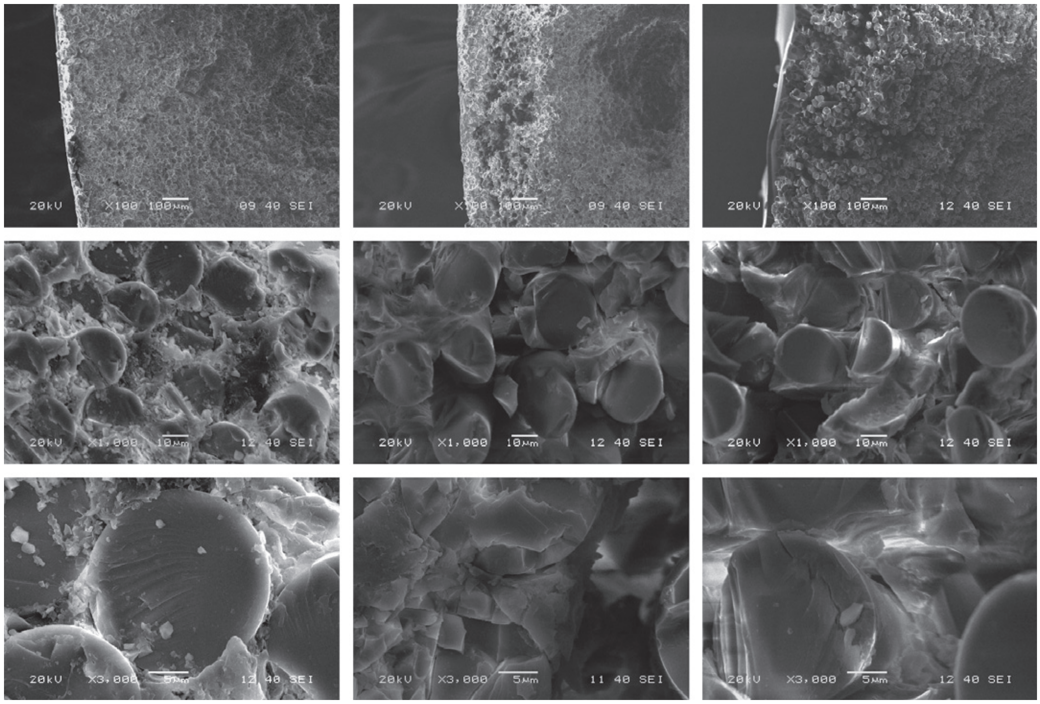


Fig. 10. Retention of tensile strains for GFRP bars after 6, 12, and 18 months in different environments.



(a) LE (b) ALK50 (c) TW50

Fig. 11. SEM micrographs of the cross sections of the bars in LE, ALK50, and TW50 environments.

and 6.9%, respectively. Similar to the degradation in the tensile strength, the maximum reductions in the fracture strains were recorded for the specimens in TW50 and ALK50 environments. These reductions were about 22.5% and 23.4%, respectively, after 18 months of exposure.

4.4. Microstructural analysis results

Fig. 11 shows the SEM micrographs of the examined GFRP samples (LE, ALK50, and TW50) at three magnifications (100×, 1000×, and 3000×). The figure at the 100× magnification shows some changes in the outer portions the GFRP bars (about 0.5–1 mm) in both the ALK50 and TW50 specimens compared to the reference specimen. At higher magnifications of 1000× and 3000×, it can be clearly seen that; while there was almost no deterioration in the glass fibers, the matrix around the glass fibers in both the ALK50 and TW50 specimens were significantly deteriorated. Many gaps were observed between fibers and at fiber-resin interfaces which affected the bond between glass fibers and vinylester resin, consequently, affected the tensile properties of the GFRP bars at failure. At lower load levels, the glass fibers in the deteriorated area withstood the applied load resulting in almost no reductions in the tensile modulus of the GFRP bars as observed from the tensile test results (reductions of only 0–5% in the tensile modulus). However, at higher load levels, close to failure, the glass fibers were weak and thus failed resulting in reductions in the tensile strength and fracture strains as observed from the test results (reductions of about 22–24% in the tensile strength and fracture strain).

4.5. Comparison between the effects of laboratory and field environments

To evaluate the effect of accelerated environmental exposures, the losses in the tensile strength of the GFRP bars exposed to different environments at the end of exposure period (after 18 months) were compared to those of the bars exposed to field environments. By dividing the tensile strength losses of the GFRP bars exposed to different laboratory environments by those of the bars exposed to field environments, an approximately equivalent exposure period of accelerated environments were computed. The equivalent exposure periods of accelerated environments to Riyadh and Gulf (Jubail) field environments are shown in Table 5.

Table 5
Equivalent exposure period of environments in comparison to Riyadh and Jubail field environments.

Environment	Strength loss (at 18 months) (%)	Equivalent period in Riyadh field		Equivalent period in Jubail field	
		$\frac{Losses_{18months}}{RF\ Loss_{18months}}$	Equivalent period* (years)	$\frac{Losses_{18months}}{RF\ Loss_{18months}}$	Equivalent period* (years)
TWR	5.36	9.6	14.3	9.0	13.5
TW50	24.48	43.7	**65.5**	41.1	**61.7**
SWR	10.71	19.1	28.7	18.0	27.0
SW50	12.80	22.8	34.2	21.5	32.2
SW50DW	9.82	17.5	26.3	16.5	24.7
ALK50	24.05	42.9	**64.3**	40.4	**60.6**
HD50	2.98	5.3	8.0	5.0	7.5
RF	0.56	1.0	1.5	0.9	1.4
JF	0.60	1.1	1.6	1.0	1.5

* Period greater than 60 years is shown in bold.

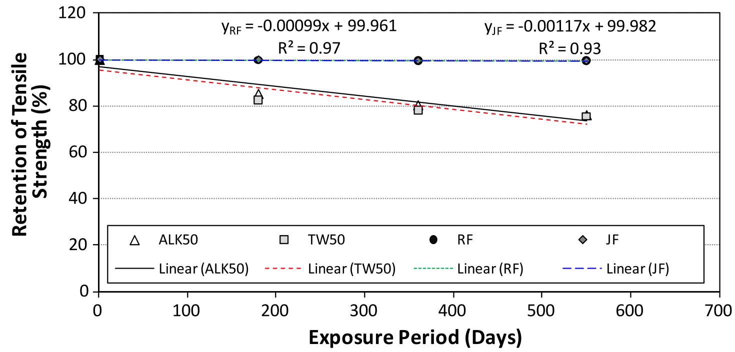


Fig. 12. Comparison between the retention of strength of GFRP bars exposed to laboratory and field conditions.

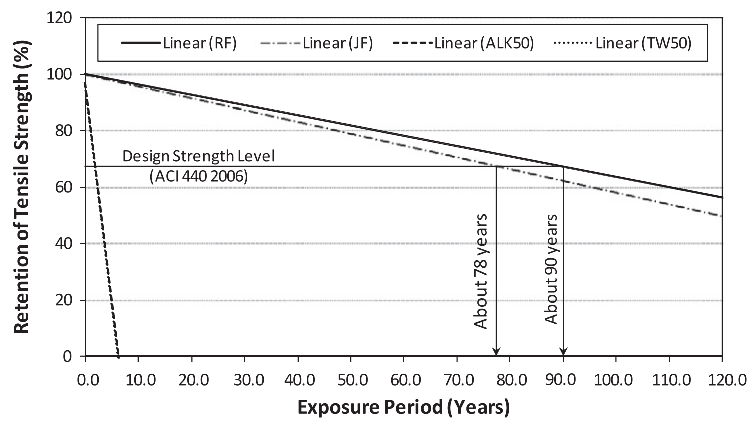


Fig. 13. Long-term predictions of the residual tensile strength of tested GFRP bars (linear regression).

Table 5 shows that 1.5 years of exposure to TWR environment was equivalent to about 14 years of exposure to Riyadh and Jubail fields; while exposure to TW50 environment was equivalent to about 66 and 62 years of exposure to Riyadh and Jubail fields, respectively. This clearly shows the effect of elevated temperature on the durability of the tested GFRP bars. The equivalent period when exposed to TW50 was more than 4 times that of the TWR environment. Exposure to ALK50 environment for 1.5 years was almost similar to the TW50 environment and was equivalent to about 64 and 61 years in Riyadh and Jubail fields, respectively.

A comparison between the two field conditions and the two most harmful accelerated laboratory environments (TW50 and ALK50) are shown in Fig. 12. Based on the test results of the two field conditions considered in this investigation, preliminary predicting equations of the tensile strength behavior of the GFRP bars were developed using a linear trend with R^2 values ranging between 0.93 and 0.97. Using these equations, the long-term prediction of the residual tensile strength of the bars in the two field conditions was extrapolated as shown in Fig. 13. It should be noted that this assumes that the loss in capacity is a linear function of time which may be conservative as the long-term shapes of many deterioration processes for residual capacity are non-linear and convex.

On the other hand, the ACI design guidelines [17] reports that the tensile strength, f_{fu}, that should be used in the design should account for the environmental exposure conditions as follows:

$$f_{fu} = C_E f_{fu}^* \tag{1}$$

where C_E is the environmental reduction factor, (0.7 for GFRP-reinforced concrete exposed to earth and weather); and f_{fu}^* is the guaranteed tensile strength of an FRP bar defined as the mean tensile strength of a sample of test specimens minus three times the standard deviation ($f_{fu}^* = f_{fu,ave} - 3\sigma$).

Using Eq. (1) and the results from Table 1, the design strength, f_{fu}, of the tested GFRP bars was computed and it was found to be about 67.2% of the average tensile strength, $f_{fu,ave}$. By considering this value in Fig. 12, it can be observed that the time when the average tensile strength to fall below the critical design value is approximately 78 and 90 years in Jubail and Riyadh field environments, respectively.

It is worth mentioning that the predicting equations shown in Fig. 12 are preliminary equations and will be improved in the future by adding additional long term results. In addition, the above mentioned equations does not consider the presence of concrete around the GFRP bars or the effects of sustained stress, both of which can affect the long-term strength of GFRP bars. These factors are currently being considered in other research studies at King Saud University, Riyadh, and their results will be presented in future publications.

5. Conclusions

Based on the results of this study, the following conclusions can be drawn:

- Regardless the type or period of exposure all tested GFRP bars had the same mode of failure and had almost linear stress–strain relationships up to failure.
- The maximum loss in the tensile strength of the tested GFRP bars was observed in the bars exposed to TW50 and ALK50 environments where the average loss was about 24.48% and 24.05%, respectively, of the initial strength after 18 months of exposure.

- For the specimens in the LE, TWR, HD laboratory environments, and the two field environments; RF and JF, almost no reduction in the tensile strength was recorded after 18 months of exposure.
- The test results showed that the GFRP bars tested in this study had better residual tensile strengths in water and alkaline solution compared to most of the GFRP bars in the literature. This shows the improvement of the new generations of GFRP bars.
- Almost no losses were observed in the tensile modulus of elasticity of the tested GFRP bars regardless of environmental exposure and exposure period; while the losses in the tensile strains were almost similar to those recorded in the tensile strength.
- The SEM results show that the matrix around the glass fibers in both ALK50 and TW50 specimens were significantly deteriorated. However, there was almost no deterioration in the glass fibers. This explains the significant losses recorded in both tensile strength and fracture strain and minor losses recorded in the tensile modulus.
- By comparing the data from laboratory and the field environments, it was observed that 1.5 years of exposure to TW50 and ALK50 environments was equivalent to about 65 and 60 years of exposure to RF and JF field conditions, respectively.
- Using a linear regression of test results, it was found that the time required for the average tensile strength in JF and RF field environments to fall below the critical design strength is approximately 75 and 90 years, respectively. However, it should be mentioned that this is a preliminary prediction and may be modified in the future when more long term results will become available.

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